Part a. Write the CRUD operation queries for the above in MongoDB NoSQL

1. Create a keyspace by name Employee

```
test> use Employee
switched to db Employee
```

2. Create a column family by name Employee-Info with attributes Emp_Id Primary Key,

Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name

```
Employee> db.createCollection ("EmployeeInfo")
{ ok: 1 }
```

3. Insert the values into the table in batch

```
Employee> db.Employee_Info.insertMany([
...     {
...       Emp_Id: 120,
...       Emp_Name: "Asha",
...       Designation: "Manager",
...       Date_of_Joining: ISODate("2021-04-01"),
...       Salary: 30000,
...       Dept_Name: "CSE"
...     },
...     {
...       Emp_Id: 121,
...       Emp_Name: "Kiran",
...       Designation: "Emp",
...       Date_of_Joining: ISODate("2019-04-20"),
...       Salary: 20000,
...       Dept_Name: "CSE"
...     },
...     {
...       Emp_Id: 122,
...       Emp_Name: "Tarun",
...       Designation: "Emp",
...       Date_of_Joining: ISODate("2019-05-01"),
...       Salary: 22000,
...       Dept_Name: "CSE"
...     },
...     {
...       Emp_Id: 123,
...       Emp_Name: "Samarth",
...       Designation: "Emp",
...       Date_of_Joining: ISODate("2020-08-01"),
...       Salary: 22500,
...       Dept_Name: "CSE"
...     },
...     {
...       Emp_Id: 124,
...       Emp_Name: "Rohan",
...       Designation: "Emp",
...       Date_of_Joining: ISODate("2019-06-01"),
...       Salary: 21000,
...       Dept_Name: "CSE"
...     }
... ])
...
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69855ca804759bf9d2628ca0'),
    '1': ObjectId('69855ca804759bf9d2628ca1'),
    '2': ObjectId('69855ca804759bf9d2628ca2'),
    '3': ObjectId('69855ca804759bf9d2628ca3'),
```

4. Update Employee name and Department of Emp-Id 121

```
Employee> db.Employee_Info.updateOne(
...     { Emp_Id: 121 },
...     { $set: { Emp_Name: "David", Dept_Name: "ECE" } }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

5. Sort the details of Employee records based on salary

```
Employee> db.Employee_Info.find().sort({ Salary: 1 })
[
  {
    _id: ObjectId('69855ca804759bf9d2628ca1'),
    Emp_Id: 121,
    Emp_Name: 'David',
    Designation: 'Emp',
    Date_of_Joining: ISODate('2019-04-20T00:00:00.000Z'),
    Salary: 20000,
    Dept_Name: 'ECE'
  },
  {
    _id: ObjectId('69855ca804759bf9d2628ca4'),
    Emp_Id: 124,
    Emp_Name: 'Rohan',
    Designation: 'Emp',
    Date_of_Joining: ISODate('2019-06-01T00:00:00.000Z'),
    Salary: 21000,
    Dept_Name: 'CSE'
  },
  {
    _id: ObjectId('69855ca804759bf9d2628ca2'),
    Emp_Id: 122,
    Emp_Name: 'Tarun',
    Designation: 'Emp',
    Date_of_Joining: ISODate('2019-05-01T00:00:00.000Z'),
    Salary: 22000,
    Dept_Name: 'CSE'
  },
  {
    _id: ObjectId('69855ca804759bf9d2628ca3'),
    Emp_Id: 123,
    Emp_Name: 'Samarth',
    Designation: 'Emp',
    Date_of_Joining: ISODate('2020-08-01T00:00:00.000Z'),
    Salary: 22500,
    Dept_Name: 'CSE'
  },
  {
    _id: ObjectId('69855ca804759bf9d2628ca0'),
    Emp_Id: 120,
    Emp_Name: 'Asha',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2021-04-01T00:00:00.000Z'),
    Salary: 30000,
    Dept_Name: 'CSE'
  }
]
```

6. Alter the schema of the table Employee_Info to add a column Projects which stores a set of

Projects done by the corresponding Employee.

```
Employee> Projects: ["Project A", "Project B"]
[ 'Project A', 'Project B' ]
Employee> db.Employee_Info.updateOne(
...   { _id: 121 },
...   {
...     $set: {
...       Projects: ["Payroll System", "Inventory App"]
...     }
...   }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

7. Update the altered table to add project names.

```
Employee> db.Employee_Info.updateOne(
...   { Emp_Id: 120 },
...   { $set: { Projects: ["Research"] } }
... )
...
... db.Employee_Info.updateOne(
...   { Emp_Id: 121 },
...   { $set: { Projects: ["Reporting"] } }
... )
...
... db.Employee_Info.updateOne(
...   { Emp_Id: 122 },
...   { $set: { Projects: ["Data Analysis"] } }
... )
...
... db.Employee_Info.updateOne(
...   { Emp_Id: 123 },
...   { $set: { Projects: ["Data Migration"] } }
... )
...
... db.Employee_Info.updateOne(
...   { Emp_Id: 124 },
...   { $set: { Projects: ["Research"] } }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

8.Create a TTL of 15 seconds to display the values of Employees.

Step 1: Add a timestamp field

Step 2: Create a TTL Index (15 seconds)

```
Employee> db.Employee_Info.updateMany(
...     {},
...     { $set: { createdAt: new Date() } }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
Employee> db.Employee_Info.createIndex(
...     { createdAt: 1 },
...     { expireAfterSeconds: 15 }
... )
...
createdAt_1
```

Step 3: View Records Before Expiry

```
Employee> db.Employee_Info.find().pretty()
[
  {
    _id: ObjectId('69855ca804759bf9d2628ca0'),
    Emp_Id: 120,
    Emp_Name: 'Asha',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2021-04-01T00:00:00.000Z'),
    Salary: 30000,
    Dept_Name: 'CSE',
    Projects: [ 'Research' ],
    createdAt: ISODate('2026-02-06T03:41:33.770Z')
  },
  {
    _id: ObjectId('69855ca804759bf9d2628ca1'),
    Emp_Id: 121,
    Emp_Name: 'David',
    Designation: 'Emp',
    Date_of_Joining: ISODate('2019-04-20T00:00:00.000Z'),
    Salary: 20000,
    Dept_Name: 'ECE',
    Projects: [ 'Reporting' ],
    createdAt: ISODate('2026-02-06T03:41:33.770Z')
  },
  {
    _id: ObjectId('69855ca804759bf9d2628ca2'),
    Emp_Id: 122,
    Emp_Name: 'Tarun',
    Designation: 'Emp',
    Date_of_Joining: ISODate('2019-05-01T00:00:00.000Z'),
    Salary: 22000,
    Dept_Name: 'CSE',
    Projects: [ 'Data Analysis' ],
    createdAt: ISODate('2026-02-06T03:41:33.770Z')
  },
  {
    _id: ObjectId('69855ca804759bf9d2628ca3'),
    Emp_Id: 123,
    Emp_Name: 'Samarth',
    Designation: 'Emp',
    Date_of_Joining: ISODate('2020-08-01T00:00:00.000Z'),
    Salary: 22500,
    Dept_Name: 'CSE',
    Projects: [ 'Data Migration' ],
    createdAt: ISODate('2026-02-06T03:41:33.770Z')
```

```
  {
    _id: ObjectId('69855ca804759bf9d2628ca3'),
    Emp_Id: 123,
    Emp_Name: 'Samarth',
    Designation: 'Emp',
    Date_of_Joining: ISODate('2020-08-01T00:00:00.000Z'),
    Salary: 22500,
    Dept_Name: 'CSE',
    Projects: [ 'Data Migration' ],
    createdAt: ISODate('2026-02-06T03:41:33.770Z')
  },
  {
    _id: ObjectId('69855ca804759bf9d2628ca4'),
    Emp_Id: 124,
    Emp_Name: 'Rohan',
    Designation: 'Emp',
    Date_of_Joining: ISODate('2019-06-01T00:00:00.000Z'),
    Salary: 21000,
    Dept_Name: 'CSE',
    Projects: [ 'Research' ],
    createdAt: ISODate('2026-02-06T03:41:33.770Z')
  }
]
```